

# $\pi^3$ : PERMUTATION-EQUIVARIANT VISUAL GEOMETRY LEARNING

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- **site:** <https://yyfz.github.io/pi3/>
- **code:** <https://github.com/yyfz/Pi3>
- **paper:** <https://arxiv.org/pdf/2507.13347>

# Why?

- mentioned in pervious papers (e.g. Easi3R, MAST3R-SLAM ...)
  - e.g. point map aligned to reference coordinate
  - is there a performance degradation due to its dependency?
- also, its weight is used for initialization for SLAM-Former

# Index

- Problem & Motivation
- Method
- Experiments
- Limitation

# Intro

- Visual geometry reconstruction

- a long-standing and fundamental problem in computer vision
- holds substantial potential for applications (e.g. AR, AN)

- methods

- traditional methods
  - use iterative optimization techniques like BA
- End-to-end models (e.g. DUS3R)
  - deep learning for reconstructing geometry from image pairs, videos, or multi-view collections

→ Despite these advances, a critical limitation persists in both classical and modern approaches: the reliance on selecting a single, fixed reference view

- problem of fixed reference view

- SfM & MVS(reference view → global coordinate)
  - introduces an unnecessary inductive bias that fundamentally constrains the performance and robustness of feed-forward neural networks
- reliance on an arbitrary reference
  - poor choice can lead to a dramatic degradation in reconstruction quality

⇒  $\pi^3$ : a robust, accurate, fully permutation-equivariant method that eliminates reference view-based biases in visual geometry learning

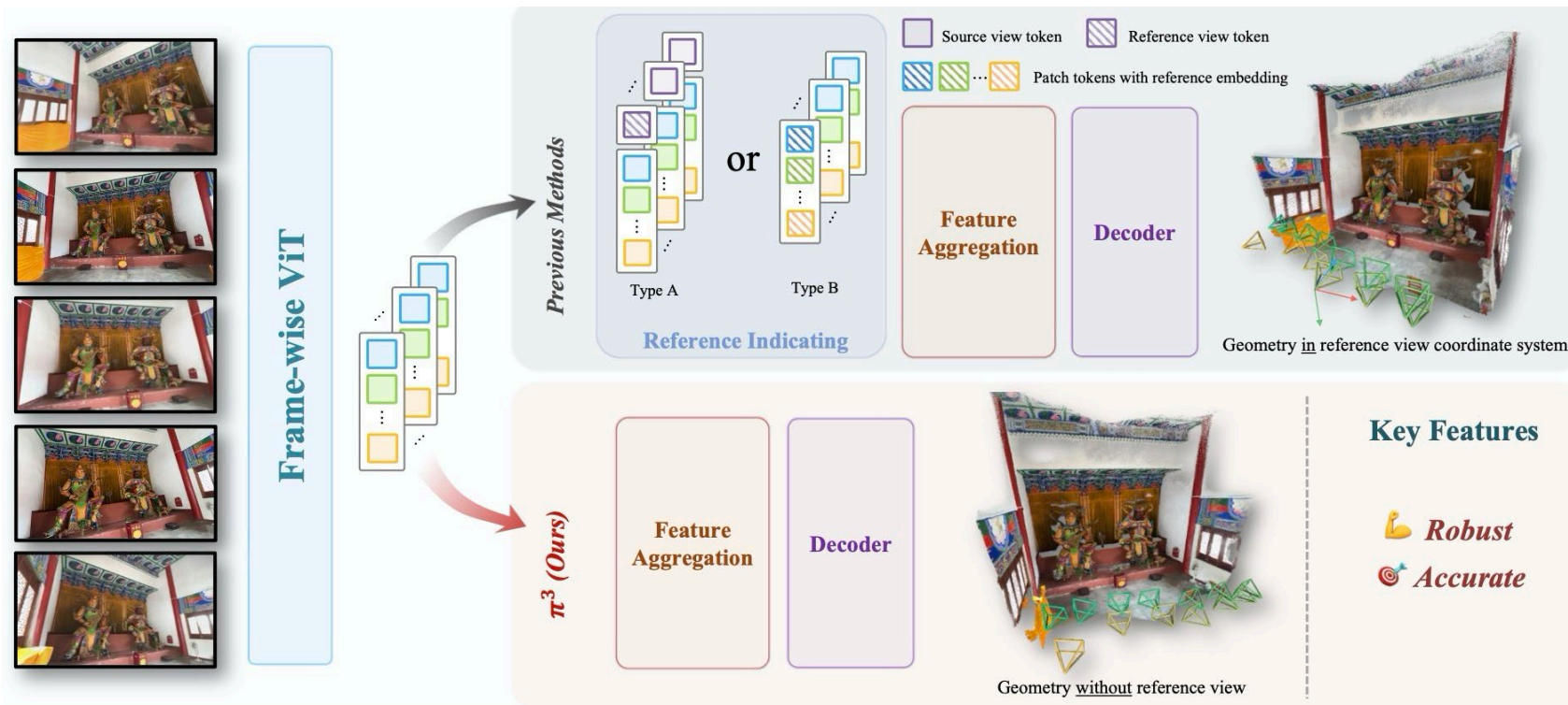
# Related Works

- Classical methods (e.g. SfM, MVS)
  - leverage the principles of multi-view geometry to establish feature correspondences across images
  - estimate camera poses and generate dense 3D point clouds
  - robust, particularly in controlled environments
    - rely on complex, multi-stage pipelines
    - involve time-consuming iterative optimization problems like BA
- feed-forward models (e.g. Dust3R)
  - focused on processing image pairs to predict a point cloud within the coordinate system of the first camera
  - effective for two views
    - scaling this to larger scenes requires a subsequent global alignment step (time-consuming & unstable)
- Subsequent work (focused on overcoming this limitation)
  - Fast3R, FLARE, VGGT
    - their reliance on anchoring the predicted 3D structure to a designated reference frame

⇒  $\pi^3$  departs from this paradigm by presenting a fundamentally different approach

$\pi^3$ 

- first to identify the reliance on a fixed reference view in visual geometry reconstruction and introduce **detrimental inductive bias** that limits model robustness and performance
- a novel, fully permutation-equivariant architecture that eliminates **this bias**
- establishes a new state-of-the-art on a wide range of benchmarks on extensive experiments



$\pi^3$ 

- network architecture ( $\phi$ )

$$\phi(S) = ((\mathbf{T}_1, \dots, \mathbf{T}_N), (\mathbf{X}_1, \dots, \mathbf{X}_N), (\mathbf{C}_1, \dots, \mathbf{C}_N)) \quad (1)$$

- **input**

- $S = (\mathbf{I}_1, \dots, \mathbf{I}_N)$   
→ sequence of  $N$  images
- where each image  $\mathbf{I}_i \in \mathbb{R}^{H \times W \times 3}$

- **output**

- $\mathbf{T}_i \in SE(3) \subset \mathbb{R}^{4 \times 4}$   
→ camera pose
- $\mathbf{X}_i \in \mathbb{R}^{H \times W \times 3}$   
→ associated pixel-aligned 3D point map (represented in its own camera coordinate system)
- $\mathbf{C}_i \in \mathbb{R}^{H \times W}$   
→ confidence map of  $\mathbf{X}_i$  each corresponding to the input image  $\mathbf{I}_i$

$\pi^3$ 

- permutation-equivariance

$$\phi(P_\pi(S)) = P_\pi(\phi(S)) \quad (2)$$

$$P_\pi(S) = (\mathbf{I}_{\pi(1)}, \dots, \mathbf{I}_{\pi(N)})$$


$$P_\pi(\phi(S)) = ((\mathbf{T}_{\pi(1)}, \dots, \mathbf{T}_{\pi(N)}), (\mathbf{X}_{\pi(1)}, \dots, \mathbf{X}_{\pi(N)}), (\mathbf{C}_{\pi(1)}, \dots, \mathbf{C}_{\pi(N)})) \quad (3)$$

- key advantages

- reconstruction quality becomes independent of the reference view selection ( $\leftrightarrow$  VGGT)
- the model becomes more robust to uncertain or noisy observations



$\pi^3$ 

- omits order-dependent components
  - positional embeddings which used to differentiate between frames
  - specialized learnable tokens which designate a reference view (e.g. camera token of VGGT)
- process
  1. embed each view into a sequence of patch tokens using a DINOv2 backbone
  2. then processed through a series of alternating **view-wise** and **global self-attention layers** (similar to VGGT) before a final decoder generates the output

# Scale-Invariant Local Geometry

- well-known challenge in monocular reconstruction is the inherent scale ambiguity

$$(\hat{\mathbf{X}}_1, \dots, \hat{\mathbf{X}}_N) \xrightarrow{S^*} (\mathbf{X}_1, \dots, \mathbf{X}_N)$$

predicted point maps                      GT point maps

- objective function

$$s^* = \arg \min_s \sum_{i=1}^N \sum_{j=1}^{H \times W} \frac{1}{z_{i,j}} \|s \hat{\mathbf{x}}_{i,j} - \mathbf{x}_{i,j}\|_1$$

- use ROE solver (proposed by MoGe)
- point cloud reconstruction loss

$$\mathcal{L}_{\text{points}} = \frac{1}{3NHW} \sum_{i=1}^N \sum_{j=1}^{H \times W} \frac{1}{z_{i,j}} \|s^* \hat{\mathbf{x}}_{i,j} - \mathbf{x}_{i,j}\|_1 \quad (5)$$

- $\hat{\mathbf{x}}_{i,j} \in \mathbb{R}^3$   
→ predicted 3D point at index  $j$  of the point map  $\hat{\mathbf{X}}_i$
- $\mathbf{x}_{i,j}$   
→ its ground-truth counterpart in  $\mathbf{X}_i$
- $z_{i,j}$   
→ ground-truth depth which is the z-component of  $\mathbf{x}_{i,j}$

# Scale-Invariant Local Geometry

- normal loss (to encourage the reconstruction of locally smooth surfaces)

$$\mathcal{L}_{\text{normal}} = \frac{1}{NHW} \sum_{i=1}^N \sum_{j=1}^{H \times W} \arccos(\hat{\mathbf{n}}_{i,j} \cdot \mathbf{n}_{i,j}) \quad (6)$$

- confidence loss (BCE)

$$\text{if } \frac{1}{z_{i,j}} \|\mathbf{s}^* \hat{\mathbf{x}}_{i,j} - \mathbf{x}_{i,j}\|_1 < \epsilon, \quad 1$$

threshold parameter

$$\text{else, } \quad 0$$

# Affine-Invariant Camera Pose

- output camera pose

$$(\hat{\mathbf{T}}_1, \dots, \hat{\mathbf{T}}_N)$$

→ only defined up to arbitrary similarity transformation

- rigid transformation
- single, unknown global scale factor

→ To resolve the ambiguity of the global reference frame, we supervise the network on the relative poses between views

- predicted relative pose

$$\hat{\mathbf{T}}_{i \leftarrow j} = \hat{\mathbf{T}}_i^{-1} \hat{\mathbf{T}}_j$$

- $\hat{\mathbf{R}}_{i \leftarrow j} \in SO(3)$
- $\hat{\mathbf{t}}_{i \leftarrow j} \in \mathbb{R}^3$

→ but the relative translation's magnitude is ambiguous

→ resolve this by leveraging the optimal scale factor  $s^*$

# Affine-Invariant Camera Pose

- camera loss

- weighted sum of a rotation loss term and a translation loss term

$$\mathcal{L}_{\text{cam}} = \frac{1}{N(N-1)} \sum_{i \neq j} (\mathcal{L}_{\text{rot}}(i, j) + \lambda_{\text{trans}} \mathcal{L}_{\text{trans}}(i, j)) \quad (8)$$

- rotation loss

$$\mathcal{L}_{\text{rot}}(i, j) = \arccos \left( \frac{\text{Tr} \left( (\mathbf{R}_{i \leftarrow j})^\top \hat{\mathbf{R}}_{i \leftarrow j} \right) - 1}{2} \right) \quad (9)$$

geodesic distance

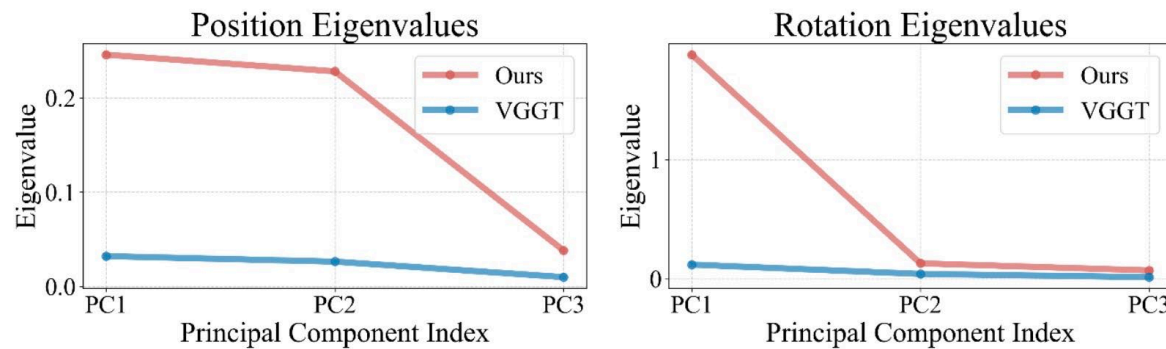
- translation loss

$$\mathcal{L}_{\text{trans}}(i, j) = \mathcal{H}_\delta(s^* \hat{\mathbf{t}}_{i \leftarrow j} - t_{i \leftarrow j}) \quad (10)$$

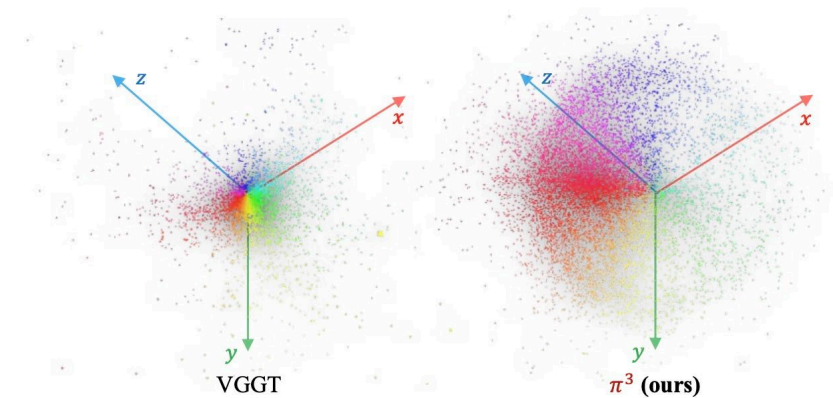
huber loss

# Affine-Invariant Camera Pose

- key insight
  - real-world camera paths are highly structured, not random
    - camera orbiting an object  $\rightarrow$  moves along a sphere
    - car-mounted camera  $\rightarrow$  follows a curve
- structure of the predicted pose



**Figure 4.** Comparison of predicted pose distributions. (eigen value analysis)



**Figure 6.** Comparison of predicted pose distributions

$\Rightarrow$  variance of our predicted poses is concentrated along significantly fewer principal components than VGGT, validating the low-dimensional structure of our output

# Model Training

- composite loss function

$$\mathcal{L} = \mathcal{L}_{\text{points}} + \lambda_{\text{normal}}\mathcal{L}_{\text{normal}} + \lambda_{\text{conf}}\mathcal{L}_{\text{conf}} + \lambda_{\text{cam}}\mathcal{L}_{\text{cam}} \quad (11)$$

- Training Datasets

- indoor, outdoor
- synthetic, real-world



15 diverse datasets

The specific datasets include GTA-SfM (Wang & Shen, 2020), CO3D (Reizenstein et al., 2021), WildRGB-D (Xia et al., 2024), Habitat (Savva et al., 2019), ARKitScenes (Baruch et al., 2021), TartanAir (Wang et al., 2020), ScanNet (Dai et al., 2017), ScanNet++ (Yeshwanth et al., 2023), BlendedMVG (Yao et al., 2020), MatrixCity (Li et al., 2023), MegaDepth (Li & Snavely, 2018), Hypersim (Roberts et al., 2021), Taskonomy (Zamir et al., 2018), Mid-Air (Fonder & Van Droogenbroeck, 2019), and an internal dynamic scene dataset. Details of model training can be found in

# Experiment

- Task

- camera pose estimation
- point map estimation
- video depth estimation
- monocular depth estimation

- Analysis

- robustness evaluation against input image sequence permutations
- ablation study on scale-invariant point maps & affine-invariant camera poses

- Training Details (appendix)

- two-stage training (similar to DUST3R)
  - **Stage 1:** 16 x A100 GPUs
  - **Stage 2:** 64 x A100 GPUs
- Dynamic Batch Sizing
  - **Stage 1:** 64 images per GPU
  - **Stage 2:** 48 images per GPU
- initialize weight(encoder, alternating attention module) with VGGT pretrained model



# Camera Pose Estimation

| Method                     | RealEstate10K |              |              | Co3Dv2 (seen) |              |              | Sintel       |              |              | TUM-dynamics |              |              | ScanNet (seen) |              |              |
|----------------------------|---------------|--------------|--------------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|----------------|--------------|--------------|
|                            | RRA ↑         | RTA ↑        | AUC ↑        | RRA ↑         | RTA ↑        | AUC ↑        | ATE ↓        | RPE-t ↓      | RPE-r ↓      | ATE ↓        | RPE-t ↓      | RPE-r ↓      | ATE ↓          | RPE-t ↓      | RPE-r ↓      |
| Fast3R (Yang et al., 2025) | 99.05         | 81.86        | 61.68        | 97.49         | 91.11        | 73.43        | 0.371        | 0.298        | 13.75        | 0.090        | 0.101        | 1.425        | 0.155          | 0.123        | 3.491        |
| CUT3R (Wang et al., 2025b) | 99.82         | 95.10        | <u>81.47</u> | 96.19         | 92.69        | 75.82        | 0.217        | 0.070        | 0.636        | 0.047        | 0.015        | 0.451        | 0.094          | 0.022        | 0.629        |
| FLARE (Zhang et al., 2025) | 99.69         | <u>95.23</u> | 80.01        | 96.38         | 93.76        | 73.99        | 0.207        | 0.090        | 3.015        | 0.026        | 0.013        | 0.475        | 0.064          | 0.023        | 0.971        |
| VGGT (Wang et al., 2025a)  | <u>99.97</u>  | 93.13        | 77.62        | <u>98.96</u>  | <u>97.13</u> | <b>88.59</b> | <u>0.167</u> | 0.062        | <u>0.491</u> | <b>0.012</b> | <u>0.010</u> | <b>0.311</b> | <u>0.035</u>   | <u>0.015</u> | <u>0.382</u> |
| $\pi^3$ (Ours)             | <b>99.99</b>  | <b>95.62</b> | <b>85.90</b> | <b>99.05</b>  | <b>97.33</b> | <u>88.41</u> | <b>0.074</b> | <b>0.040</b> | <b>0.282</b> | <u>0.014</u> | <b>0.009</b> | <u>0.312</u> | <b>0.031</b>   | <b>0.013</b> | <b>0.347</b> |

Table 1. Camera pose estimation.

## Metric & Datasets

- angular accuracy (RRA, RTA, AUC)  
→ RealEstate10K, Co3Dv2
- distance error (ATE, RPE-t, RPE-r)  
→ Sintel, TUM-dynamics, ScanNet

## Result

- new SOTA benchmark in zero-shot generalization  
→ Sintel, RealEstate10K
- achieves competitive SOTA results alongside VGGT on  
→ TUM-dynamics, in-domain Co3Dv2, ScanNet datasets

⇒  $\pi^3$  shows strong generalization capabilities while maintaining excellent performance on familiar data distributions

# Point Map Estimation

## Datasets

- 7-Scenes, NRGBD (sparse & dense)
- DTU, ETH3D

## Metric

- Accuracy (Acc.)
- Completion (Comp.)
- Normal Consistency (N.C.)

## Process

- Sim(3) alignment using Umeyama algorithm
- refinement with the ICP algorithm

| Method                     | View   | 7-Scenes     |              |              |              |              |              | NRGBD        |              |              |              |              |              |
|----------------------------|--------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                            |        | Acc. ↓       |              | Comp. ↓      |              | NC. ↑        |              | Acc. ↓       |              | Comp. ↓      |              | NC. ↑        |              |
|                            |        | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         |
| Fast3R (Yang et al., 2025) | sparse | 0.095        | 0.065        | 0.144        | 0.089        | 0.673        | 0.759        | 0.135        | 0.091        | 0.163        | 0.104        | 0.759        | 0.877        |
| CUT3R (Wang et al., 2025b) |        | 0.093        | 0.049        | 0.102        | 0.051        | 0.704        | 0.805        | 0.104        | 0.041        | 0.079        | 0.031        | 0.822        | 0.968        |
| FLARE (Zhang et al., 2025) |        | 0.085        | 0.057        | 0.145        | 0.107        | 0.696        | 0.780        | 0.053        | <u>0.024</u> | <u>0.051</u> | <u>0.025</u> | 0.877        | <u>0.988</u> |
| VGGT (Wang et al., 2025a)  |        | <b>0.044</b> | <b>0.025</b> | <b>0.056</b> | <b>0.033</b> | <u>0.733</u> | <b>0.845</b> | <u>0.051</u> | 0.029        | 0.066        | 0.038        | <u>0.890</u> | 0.981        |
| $\pi^3$ (Ours)             |        | 0.047        | 0.029        | 0.075        | 0.049        | <b>0.742</b> | 0.841        | <b>0.026</b> | <b>0.015</b> | <b>0.028</b> | <b>0.014</b> | <b>0.916</b> | <b>0.992</b> |
| Fast3R (Yang et al., 2025) | dense  | 0.040        | 0.017        | 0.056        | 0.018        | 0.644        | 0.725        | 0.072        | 0.030        | 0.050        | 0.016        | 0.790        | 0.934        |
| CUT3R (Wang et al., 2025b) |        | 0.023        | 0.010        | <u>0.027</u> | <b>0.008</b> | 0.669        | 0.764        | 0.086        | 0.037        | 0.048        | 0.017        | 0.800        | 0.953        |
| FLARE (Zhang et al., 2025) |        | <u>0.019</u> | <u>0.007</u> | 0.026        | 0.013        | <u>0.684</u> | 0.785        | 0.023        | 0.011        | 0.018        | 0.008        | 0.882        | 0.986        |
| VGGT (Wang et al., 2025a)  |        | 0.022        | 0.008        | 0.026        | 0.012        | 0.666        | 0.760        | <u>0.017</u> | <u>0.010</u> | <u>0.015</u> | <u>0.005</u> | <u>0.893</u> | <b>0.988</b> |
| $\pi^3$ (Ours)             |        | <b>0.016</b> | <b>0.007</b> | <b>0.022</b> | <u>0.011</u> | <b>0.689</b> | <b>0.792</b> | <b>0.015</b> | <b>0.008</b> | <b>0.013</b> | <b>0.005</b> | <b>0.898</b> | <u>0.987</u> |

Table 2. Point map estimation on 7-Scenes and NRGBD

| Method                     | DTU          |              |              |              |              |              | ETH3D        |              |              |              |              |              |
|----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                            | Acc. ↓       |              | Comp. ↓      |              | N.C. ↑       |              | Acc. ↓       |              | Comp. ↓      |              | N.C. ↑       |              |
|                            | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         |
| Fast3R (Yang et al., 2025) | 3.340        | 1.919        | 2.929        | 1.125        | 0.671        | 0.755        | 0.832        | 0.691        | 0.978        | 0.683        | 0.667        | 0.766        |
| CUT3R (Wang et al., 2025b) | 4.742        | 2.600        | 3.400        | 1.316        | <u>0.679</u> | 0.764        | 0.617        | 0.525        | 0.747        | 0.579        | 0.754        | 0.848        |
| FLARE (Zhang et al., 2025) | 2.541        | 1.468        | 3.174        | 1.420        | <b>0.684</b> | <b>0.774</b> | 0.464        | 0.338        | 0.664        | 0.395        | 0.744        | 0.864        |
| VGGT (Wang et al., 2025a)  | 1.338        | 0.779        | 1.896        | 0.992        | 0.676        | 0.766        | 0.280        | <u>0.185</u> | <u>0.305</u> | <u>0.182</u> | <u>0.853</u> | <u>0.950</u> |
| $\pi^3$ (Ours)             | <b>1.198</b> | <b>0.646</b> | <b>1.849</b> | <b>0.607</b> | 0.678        | <u>0.768</u> | <b>0.194</b> | <b>0.131</b> | <b>0.210</b> | <b>0.128</b> | <b>0.883</b> | <b>0.969</b> |

Table 3. Point map estimation on DTU and ETH3D

## Result

- strong generalization capability in a broad spectrum of 3D reconstruction tasks
- robust across synthetic and real-world scenarios, sparse and dense view settings (Table 2), as well as object-level and scene-level scales (Table 3)

# Depth Estimation

## Datasets

- Sintel
- Bonn
- KITTI
- NYU-v2 (only in monocular depth estimation)

## Metric

- $\delta < 1.25$
- Abs Rel

trained with more  
synthetic & driving data

## Result (Video depth estimation)

- new SOTA performance across all three datasets within feed-forward 3D reconstruction methods
- running at 57.4 FPS on KITTI
  - significantly faster than VGGT and Aether despite having a smaller model size

| Method                           | Params | Sintel       |                          | Bonn         |                          | KITTI        |                          | FPS         |
|----------------------------------|--------|--------------|--------------------------|--------------|--------------------------|--------------|--------------------------|-------------|
|                                  |        | Abs Rel ↓    | $\delta < 1.25 \uparrow$ | Abs Rel ↓    | $\delta < 1.25 \uparrow$ | Abs Rel ↓    | $\delta < 1.25 \uparrow$ |             |
| DUST3R (Wang et al., 2024)       | 571M   | 0.662        | 0.434                    | 0.151        | 0.839                    | 0.143        | 0.814                    | 1.25        |
| MASt3R (Leroy et al., 2024)      | 689M   | 0.558        | 0.487                    | 0.188        | 0.765                    | 0.115        | 0.848                    | 1.01        |
| MonST3R (Zhang et al., 2024)     | 571M   | 0.399        | 0.519                    | 0.072        | 0.957                    | 0.107        | 0.884                    | 1.27        |
| Fast3R (Yang et al., 2025)       | 648M   | 0.638        | 0.422                    | 0.194        | 0.772                    | 0.138        | 0.834                    | <b>65.8</b> |
| MVDUST3R (Tang et al., 2024)     | 661M   | 0.805        | 0.283                    | 0.426        | 0.357                    | 0.456        | 0.342                    | 0.69        |
| CUT3R (Wang et al., 2025b)       | 793M   | 0.417        | 0.507                    | 0.078        | 0.937                    | 0.122        | 0.876                    | 6.98        |
| Aether (Team et al., 2025)       | 5.57B  | 0.324        | 0.502                    | 0.273        | 0.594                    | <u>0.056</u> | <u>0.978</u>             | 6.14        |
| FLARE (Zhang et al., 2025)       | 1.40B  | 0.729        | 0.336                    | 0.152        | 0.790                    | 0.356        | 0.570                    | 1.75        |
| VGGT (Wang et al., 2025a)        | 1.26B  | 0.299        | 0.638                    | 0.057        | 0.966                    | 0.062        | 0.969                    | 43.2        |
| <b><math>\pi^3</math> (Ours)</b> | 959M   | <b>0.233</b> | <b>0.664</b>             | <b>0.049</b> | <b>0.975</b>             | <b>0.038</b> | <b>0.986</b>             | <u>57.4</u> |

Table 4. Video depth estimation on Sintel, Bonn and KITTI.

| Method                           | Sintel       |                          | Bonn         |                          | KITTI        |                          | NYU-v2       |                          |
|----------------------------------|--------------|--------------------------|--------------|--------------------------|--------------|--------------------------|--------------|--------------------------|
|                                  | Abs Rel↓     | $\delta < 1.25 \uparrow$ | Abs Rel↓     | $\delta < 1.25 \uparrow$ | Abs Rel↓     | $\delta < 1.25 \uparrow$ | Abs Rel↓     | $\delta < 1.25 \uparrow$ |
| DUST3R (Wang et al., 2024)       | 0.488        | 0.532                    | 0.139        | 0.832                    | 0.109        | 0.873                    | 0.081        | 0.909                    |
| MASt3R (Leroy et al., 2024)      | 0.413        | 0.569                    | 0.123        | 0.833                    | 0.077        | 0.948                    | 0.110        | 0.865                    |
| MonST3R (Zhang et al., 2024)     | 0.402        | 0.525                    | 0.069        | 0.954                    | 0.098        | 0.895                    | 0.094        | 0.887                    |
| Fast3R (Yang et al., 2025)       | 0.544        | 0.509                    | 0.169        | 0.796                    | 0.120        | 0.861                    | 0.093        | 0.898                    |
| CUT3R (Wang et al., 2025b)       | 0.418        | 0.520                    | 0.058        | 0.967                    | 0.097        | 0.914                    | 0.081        | 0.914                    |
| FLARE (Zhang et al., 2025)       | 0.606        | 0.402                    | 0.130        | 0.836                    | 0.312        | 0.513                    | 0.089        | 0.898                    |
| VGGT (Wang et al., 2025a)        | 0.335        | 0.599                    | 0.053        | 0.970                    | 0.082        | 0.947                    | 0.056        | 0.951                    |
| MoGe                             | <b>0.273</b> | <b>0.695</b>             | <u>0.050</u> | <u>0.976</u>             | <b>0.049</b> | <b>0.979</b>             | <u>0.055</u> | <u>0.952</u>             |
| - v1 (Wang et al., 2025c)        | - 0.273      | - 0.695                  | - 0.050      | - 0.976                  | - 0.054      | - 0.977                  | - 0.055      | - 0.952                  |
| - v2 (Wang et al., 2025d)        | - 0.277      | - 0.687                  | - 0.063      | - 0.973                  | - 0.049      | - 0.979                  | - 0.060      | - 0.940                  |
| <b><math>\pi^3</math> (Ours)</b> | <u>0.277</u> | <u>0.614</u>             | <b>0.044</b> | <b>0.976</b>             | <u>0.060</u> | <u>0.971</u>             | <b>0.054</b> | <b>0.956</b>             |

Table 5. Monocular depth estimation

## Result (Monocular depth estimation)

- SoTA results among multi-frame feed-forward reconstruction approaches, even though it is not explicitly optimized for single-frame depth estimation
- competitively with MoGe, one of the top-performing monocular depth estimation models

# Robustness Estimation

| Method                     | DTU          |              |              |              |              |              | ETH3D        |              |              |              |              |              |
|----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                            | Acc. std. ↓  |              | Comp. std. ↓ |              | N.C. std. ↓  |              | Acc. std. ↓  |              | Comp. std. ↓ |              | N.C. std. ↓  |              |
|                            | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         |
| Fast3R (Yang et al., 2025) | 0.578        | 0.451        | 0.677        | 0.376        | 0.007        | 0.009        | 0.182        | 0.205        | 0.381        | 0.273        | 0.047        | 0.072        |
| FLARE (Zhang et al., 2025) | 0.720        | 0.494        | 1.346        | 1.134        | 0.009        | 0.012        | 0.171        | 0.187        | 0.251        | 0.188        | 0.048        | 0.053        |
| VGGT (Wang et al., 2025a)  | <u>0.033</u> | <u>0.022</u> | <u>0.054</u> | <u>0.036</u> | <u>0.007</u> | <u>0.007</u> | <u>0.049</u> | <u>0.040</u> | <u>0.062</u> | <u>0.042</u> | <u>0.022</u> | <u>0.015</u> |
| $\pi^3$ (Ours)             | <b>0.003</b> | <b>0.002</b> | <b>0.006</b> | <b>0.003</b> | <b>0.001</b> | <b>0.001</b> | <b>0.000</b> | <b>0.000</b> | <b>0.000</b> | <b>0.000</b> | <b>0.001</b> | <b>0.000</b> |

Table 6. Standard deviation of point cloud estimation

## Datasets

- DTU
- ETH3D

## Metric

- Acc. std, Comp. std, N.C. std
  - mean, median

## Result

- **near-zero standard deviation** across all metrics on DTU and ETH3D, outperforming existing approaches
  - highlights the **limitations of reference-frame-dependent methods** which exhibit **significant sensitivity to input order**
- provide compelling evidence that the proposed architecture is genuinely permutation-equivariant
  - ensuring **consistent and order-independent 3D reconstruction**

lower standard deviation = higher robustness to input order variations

# Ablation Study

| Model      | ETH3D        |              |              |              |              |              | 7-Scenes     |              |              |              |              |              | NRGBD        |              |              |              |              |              |
|------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|            | Acc. ↓       |              | Comp. ↓      |              | N.C ↑        |              | Acc. ↓       |              | Comp. ↓      |              | N.C ↑        |              | Acc. ↓       |              | Comp. ↓      |              | N.C ↑        |              |
|            | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         | Mean         | Med.         |
| Model 1    | 0.229        | 0.150        | 0.166        | 0.103        | 0.802        | 0.930        | <u>0.020</u> | <u>0.010</u> | <b>0.019</b> | <u>0.009</u> | 0.715        | 0.834        | 0.034        | <u>0.018</u> | 0.025        | <u>0.011</u> | 0.859        | 0.977        |
| Model 2    | <u>0.197</u> | <u>0.118</u> | <u>0.118</u> | <u>0.065</u> | <u>0.820</u> | <u>0.943</u> | <u>0.020</u> | <b>0.009</b> | <u>0.020</u> | <b>0.008</b> | <u>0.716</u> | <u>0.837</u> | <u>0.031</u> | <u>0.018</u> | <u>0.023</u> | <b>0.010</b> | <u>0.861</u> | <u>0.978</u> |
| Full Model | <b>0.131</b> | <b>0.076</b> | <b>0.079</b> | <b>0.043</b> | <b>0.841</b> | <b>0.957</b> | <b>0.019</b> | <b>0.009</b> | <u>0.020</u> | <u>0.009</u> | <b>0.723</b> | <b>0.843</b> | <b>0.028</b> | <b>0.015</b> | <b>0.022</b> | <b>0.010</b> | <b>0.875</b> | <b>0.981</b> |

Table 7. Ablation study on the key components of our model.

## Datasets

- ETH3D (outdoor)
- 7-Scenes (indoor)
- NRGBD (indoor)

## Model

- **Model 1:** no affine-invariant poses & scale-invariant pointmaps
- **Model 2:** no affine-invariant camera pose modeling
- **Full Model:**  $\pi^3$

## Metric

- same as Point Map Estimation

## Result

- scale-invariant pointmap modeling
  - significant performance gains on outdoor datasets
  - but not indoor datasets
- outdoor scenes are more significantly affected by scale ambiguity
- affine-invariant + scale-invariant permutation-equivariant

# Conclusion

- $\pi^3$ 
  - a feed-forward neural network that presents a new paradigm for visual geometry reconstruction by [eliminating the reliance on a fixed reference view](#)
  - leverage a [fully permutation-equivariant architecture](#)
    - inherently robust to input ordering and leads to higher accuracy
  - This design choice [removes a critical inductive bias](#) found in previous methods
    - achieve state-of-the-art performance on a wide array of tasks

## Limitation

- unable to handle transparent objects ( $\pi^3$  does not explicitly account for complex light transport phenomena)
- lacks the same level of fine-grained detail (compared to contemporary diffusion-based approaches)
- point cloud generation relies on a simple upsampling mechanism using an MLP with pixel shuffling
  - [efficient, but can introduce noticeable grid-like artifacts, particularly in regions with high reconstruction uncertainty](#)

# Why Good?

- $\pi^3$ 
  - benefits from pretrained VGGT
  - consider scale
  - all-pair relative pose supervision